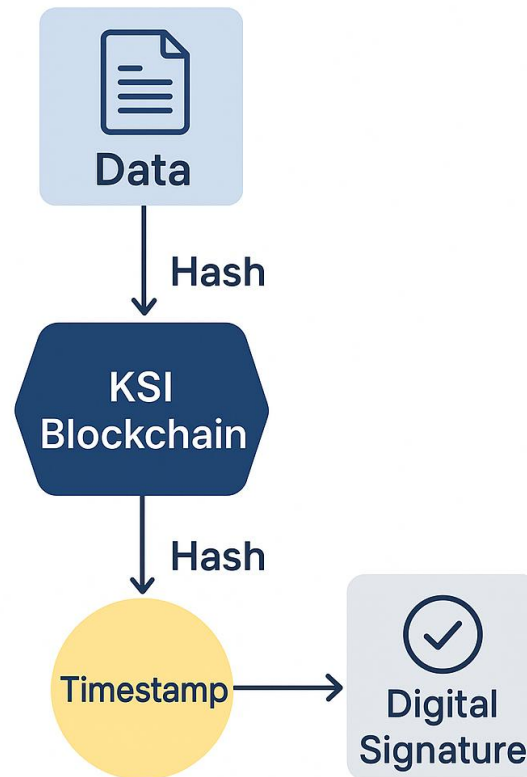


KSI Blockchain of Estonia

Securing Digital Governance through Blockchain Technology



Presented by:

Dipa Oli

Roll no: 15

BPSG 7th Sem

Understanding KSI Blockchain

- KSI (Keyless Signature Infrastructure) is a blockchain-based cybersecurity technology.
 - Unlike cryptocurrency blockchains, KSI does not store money or transactions.
 - It ensures data integrity by mathematically proving that data has not been altered.
 - Uses cryptographic hash functions instead of private keys.
 - Every data change is time-stamped and verifiable by anyone.
 - Adopted nationally by Estonia to protect government databases.

Why Estonia Implemented KSI Blockchain?

- Estonia is a highly digital state with over 99% of public services online.
 - In 2007, Estonia faced large-scale cyberattacks targeting government infrastructure.
 - Need to protect sensitive data such as health records, court files, and population data.
 - Maintaining citizen trust in e-governance was a national priority.
 - KSI provided a cost-effective, scalable, and highly secure solution.
 - It allowed Estonia to become a global leader in digital governance.

Impact of KSI on Estonian Government Services

- Secures national databases including healthcare, judiciary, and land registry.
 - Enables transparent auditing of government actions.
 - Prevents unauthorized data manipulation, even by insiders.
 - Supports digital ID, e-Residency, and paperless governance.
 - Reduces bureaucracy, service delivery time, and administrative costs.
 - Strengthens public confidence in digital government services.

Lessons and Applications for Nepal

- Nepal can apply blockchain to land records, citizenship, and public procurement.
 - Helps reduce corruption by ensuring transparency and traceability of records.
 - Protects government databases from cyber threats and unauthorized changes.
 - Can strengthen trust in digital services under the Digital Nepal Framework.
 - Adoption should be gradual, starting with high-impact sectors.
 - Requires legal reforms, digital literacy, and institutional capacity building.



Thank You